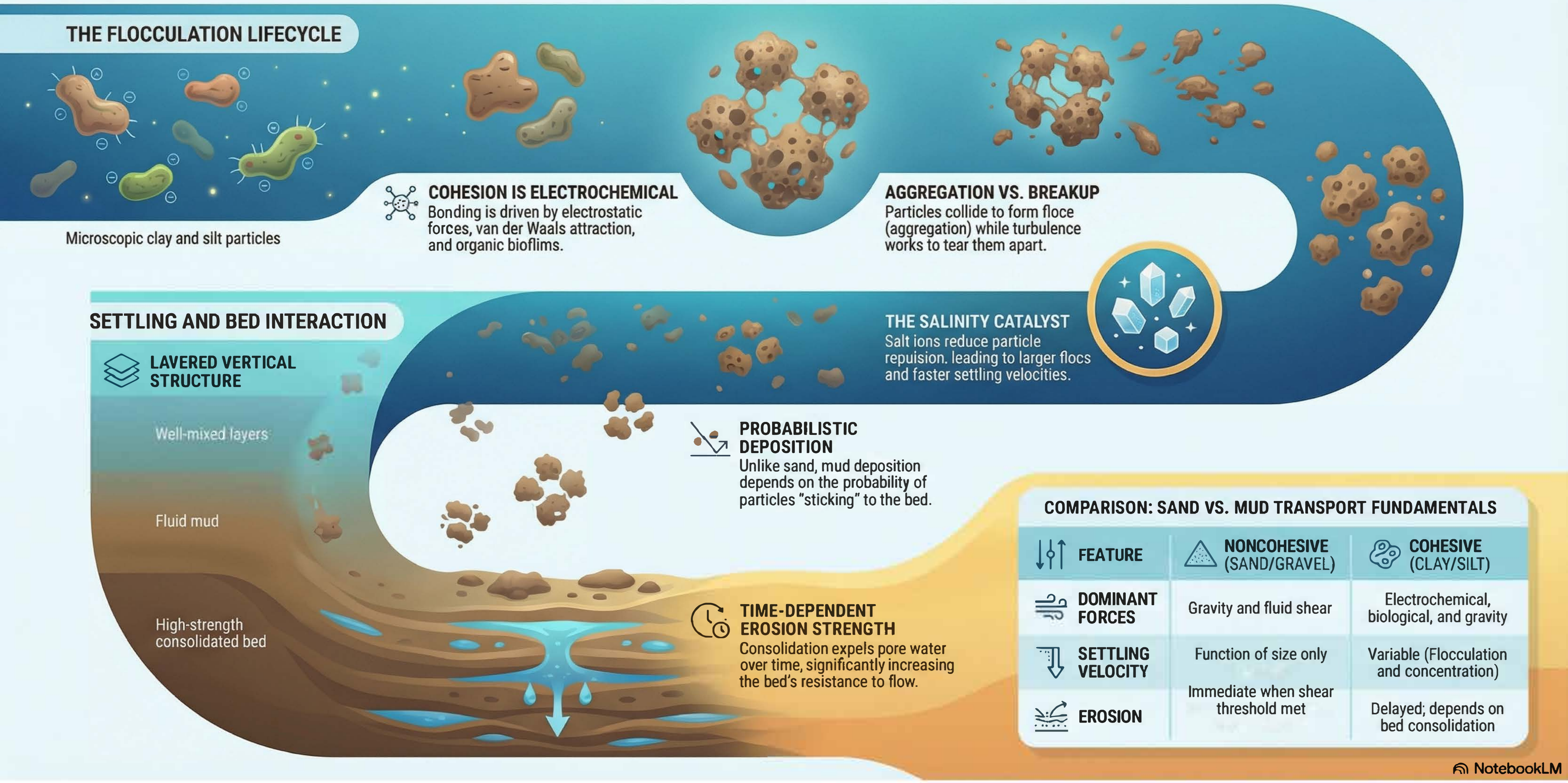
The Science of Mud: Dynamics of Cohesive Sediment Transport



COMPARISON: SAND VS. MUD TRANSPORT FUNDAMENTALS

FEATURE	NONCOHESIVE (SAND/GRAVEL)	COHESIVE (CLAY/SILT)
DOMINANT FORCES	Gravity and fluid shear	Electrochemical, biological, and gravity
SETTLING VELOCITY	Function of size only	Variable (Flocculation and concentration)
EROSION	Immediate when shear threshold met	Delayed; depends on bed consolidation